

Preparing for a Pandemic: Accelerating Vaccine Availability

Executive Summary

Governments can improve the likelihood and timeliness of pandemic-time vaccine availability by funding manufacturing capacity for a broad portfolio of vaccine candidates in parallel with their development.

Summary

Timely vaccine rollout during a pandemic has enormous social benefits, but is not guaranteed by market and regulatory forces. Social and political limits on acceptable vaccine prices mean that the social value of vaccines far exceeds their private returns.^[1] Furthermore, vaccine developers typically delay investments in manufacturing capacity until vaccine candidates' technical and regulatory uncertainty are resolved. In light of these inefficiencies, Ahuja et al. provides a succinct discussion of how governments can most effectively accelerate pandemic-time vaccine availability.

Key Point 1: Fund a diverse vaccine portfolio to increase the odds of vaccine delivery.

Developing vaccines for novel pandemic pathogens involves working at the scientific frontier where vaccine effectiveness and safety are uncertain. Governments should therefore support a diversified portfolio of vaccine candidates and technologies that reduce the total and correlated risk of failure.^[2] A diverse portfolio of candidates that succeed also reduces the risk of shortages and monopolies that incentivize vaccine nationalism.

Key Point 2: Use push funding to incentivize parallelized vaccine development and manufacturing capacity installation.

Governments can speed up vaccine availability by de-risking investments in manufacturing capacity made in parallel with vaccine development. To do so, governments can either:

1. Directly compensate firms for upfront investments in manufacturing capacity and commit to purchasing their vaccines at marginal cost (**push funding**), or
2. Commit to purchasing a given supply of vaccines at a rate that justifies early manufacturing investments given firms' beliefs that their vaccine R&D will succeed (**pull funding**).

Based on a stylized model of vaccine contracting under incomplete information, Ahuja et al. recommends contracts primarily based on push funding, as it is likely to lead to lower

government expense. If a government wants many firms to attempt vaccine development but cannot distinguish between their likelihood of success, pull funding may result in exorbitant payments to incentivize firms with lower success probability. At the same time, governments will not want to spend on manufacturing capacity for firms that are very unlikely to succeed, and will want to incentivize speed in vaccine development, too. Contracts should therefore include some pull funding.

Key Point 3: Invest in substantial manufacturing capacity, and do so early.

If vaccine supply is bottlenecked by manufacturing capacity, prices will skyrocket and shortages may ensue. During COVID-19 pandemic, firms were able to expand capacity more quickly than expected, but even greater capacity would have allowed the world to complete vaccination one year earlier than it did. Timely expansions in manufacturing capacity can be incentivized using pull financing as discussed in point 2, and can be further enabled by subsidizing the supply of critical materials like bioreactors, delivery devices, and adjuvants so they are available when a pandemic breaks out.

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1. Castillo et al. (2021) estimates that increasing vaccine capacity from 2 to 3 billion doses per year would have generated \$2.5 trillion in social value, while resulting in only \$30 billion in additional firm revenue at \$15 per dose.↵
 2. A nation's optimal vaccine portfolio does depend on its income. High income countries will want to support a large portfolio, while less wealthy countries will want to focus on select candidates that are most likely to succeed.↵